1

Environment and Ecology

1.1 INTRODUCTION

'Environment' is a term derived from the French word 'Environner' that means 'to surround'. There was a time when environment just meant surroundings. It was used to describe the physical world surrounding us including soil, rocks, water and air. Gradually it was realized that the enormous variety of plants, animals and micro-organisms on this earth, including human beings are an integral part of the environment. Hence, to make a sensible definition of environment, it was necessary to include the interactions and inter-relationships of all living organisms with the physical surroundings.

Later, it was further recognised that all types of social, cultural and technological activities carried out by human beings also have a profound influence on various components of the environment. Thus various built structures, materials and technological innovations also became a part of the environment. So now all biological (biotic) and non-biological (abiotic) entities surrounding us are included in the term 'environment'. The impact of technological and economic development on the natural environment may lead to degradation of the social and cultural environment. Thus, environment is to be considered in a broader perspective where the surrounding components as well as their interactions are to be included.

As per Environment (Protection) Act, 1986, environment includes all the physical and biological surroundings of an organism along with their interactions. **Environment** is thus defined as **"the sum total of water, air and land and the inter-relationships that exist among them and with the human beings, other living organisms and materials."** The concept of environment can be clearly understood from Fig. 1.1.

Figure 1.1 depicts the environment of human beings. Air, water and land surrounding us constitute our environment, and influence us directly. At the same time we too have an influence on our environment due to overuse or over-exploitation of resources or due to discharge of pollutants in the air, water and land. The flora, fauna and micro-organisms as well as the man-made structures in our surroundings have a bi-directional interaction with us directly or indirectly. The totality of all these components and their interactions constitute the environment.

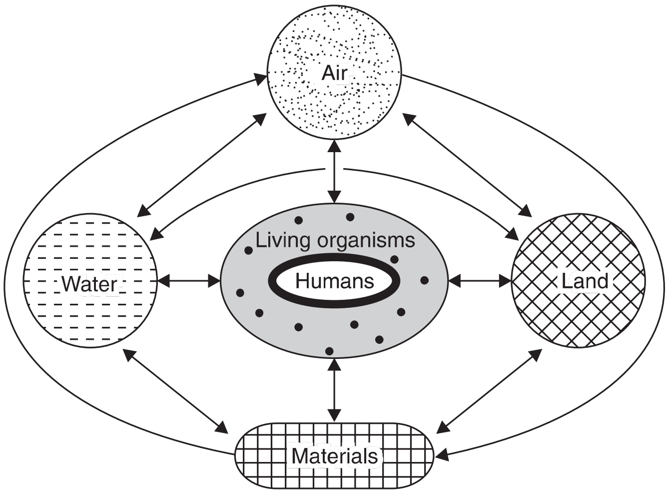


Fig. 1.1. Concept of Environment: air, water, land, living organisms and materials surrounding us and their interactions together constitute environment.

Urban environment is somewhat different from rural environment. In urban environment we can see profound influence of human beings. Most of the natural landscapes in cities have been changed and modified by man-made artificial structures like multi-storeyed buildings, commercial complexes, factories, transportation networks and so on. Urban air, water and soil are loaded with various types of chemicals and wastes. Diversity of plants and animals is much less as compared to rural environment. Urban population is more dense and has greater energy demands.

1.2 SCOPE

Environmental studies as a subject has a wide scope. It encompasses a large number of areas and aspects, which may be summarized as follows:

- Natural resources—their conservation and management
- Ecology and biodiversity
- Environmental pollution and control
- Social issues in relation to development and environment
- Human population and environment

These are the basic aspects of environmental studies which have a direct relevance to every section of the society. Environmental studies can also be highly specialized concentrating on more technical aspects like environmental science, environmental engineering or environmental management.

In the recent years, the scope of environmental studies has expanded dramatically the world over. Several career options have emerged in this field that are broadly categorized as:

(*i*) **Research & Development (R & D) in environment:** Skilled environmental scientists have an important role to play in examining various environmental problems in a scientific manner and carry out R & D activities for developing cleaner technologies and promoting sustainable development.

There is a need for trained manpower at every level to deal with environmental issues. Environmental management and environmental engineering are emerging as new career opportunities for environmental protection and management. With the pollution control laws becoming more stringent, industries are finding it difficult to dispose off the wastes produced. In order to avoid expensive litigation, companies are now trying to adopt green technologies, which would reduce pollution.

Investing in pollution control technologies will reduce pollution as well as cut on costs for effluent treatment. Market for pollution control technology is increasing the world over. Cleaning up of the wastes produced is another potential market. It is estimated to be more than $ 100 billion per year for all American business. Germany and Japan having more stringent laws for many years have gained more experience in reducing effluents. Still there is a $ 200 billion market for cleaning up the former East Germany alone. In India also the Pollution Control Boards are seriously implementing pollution control laws and insisting on upgradation of effluents to meet the prescribed standards before they are discharged on land or into a water body. Many companies not complying with the orders have been closed or ordered to shift.

(*ii*) **Green advocacy:** With increasing emphasis on implementing various Acts and Laws related to environment, need for environmental lawyers has emerged, who should be able to plead the cases related to water and air pollution, forest, wildlife etc.

(*iii*) **Green marketing:** While ensuring the quality of products with ISO mark, now there is an increasing emphasis on marketing goods that are environment friendly. Such products have ecomark or ISO 14000 certification. Environmental auditors and environmental managers would be in great demand in the coming years.

(*iv*) **Green media:** Environmental awareness can be spread amongst masses through mass media like television, radio, newspaper, magazines, hoardings, advertisements etc. for which environmentally educated persons are required.

(*v*) **Environment consultancy:** Many non-government organisations (NGOs), industries and government bodies are engaging environmental consultants for systematically studying and tackling environment related problems.

1.3 IMPORTANCE OF ENVIRONMENT

Environment belongs to all and is important to all. Whatever be the occupation or age of a person, he will be affected by environment and also he will affect the environment by his deeds. That is why we find an internationally observed environment calender to mark some important aspect or issue of environment.

Environment Calender

World Wetland Day	February 2
World Forest Day	March 21
World Day for Water	March 22
World Meteorological Day	March 23
Earth Day	April 22
International Biodiversity Day	May 22
Anti-tobacco Day	May 31
World Environment Day	June 5
World Ocean Day	June 8
World Population Day	July 11
Ozone Week	Sept. 16–23
World Car-free Day	Sept. 22
Green Consumer Day	Sept. 28
World Farm Animal's Day	Oct. 2
World Habitat Day	Oct. 3
World Animal Welfare Day	Oct. 4
Wildlife Week	Oct. 1–7
International Day for Natural Disaster Reduction	Oct. 13
World Conservation Day	Oct. 24
International Day for Biological Diversity	Dec. 29

(*a*) **Global vs. Local Importance of Environment**

Environment is one subject that is actually global as well as local in nature.

Issues like global warming, depletion of ozone layer, dwindling forests and energy resources, loss of global biodiversity etc. which are going to affect the mankind as a whole are global in nature and for that we have to think and plan globally.

However, there are some environmental problems which are of localized importance. For dealing with local environmental issues, e.g. impact of mining or hydroelectric project in an area, problems of disposal and management of solid waste, river or lake pollution, soil erosion, water logging and salinization of soil, fluorosis problem in local population, arsenic pollution of groundwater etc., we have to think and act locally.

In order to make people aware about those aspects of environment with which they are so intimately associated, it is very important to make every one environmentally educated.

(*b*) **Individualistic Importance of Environment**

Environmental studies is very important since it deals with the most mundane problems of life where each individual matters, like dealing with safe and clean drinking water, hygienic living conditions, clean and fresh air, fertile land, healthy food and sustainable development. If we want to live in a clean, healthy, aesthetically beautiful, safe and secure environment for a long time and wish to hand over a clean and safe earth to our children, grandchildren and great grandchildren, it is most essential to understand the basics of environment.

1.4 NEED FOR PUBLIC AWARENESS

(*a*) **International Efforts for Environment**

Environmental issues received international attention about 35 years back in Stockholm Conference, held on 5th June, 1972. Since then we celebrate **World Environment Day** on **5th June.** At the United Nations Conference on **Environment and Development** held at Rio de Janeiro, in 1992, known popularly as **Earth Summit,** and ten years later, the **World Summit on Sustainable Development,** held at Johannesberg in 2002, key issues of global environmental concern were highlighted. Attention of general public was drawn towards the deteriorating environmental conditions all over the world.

Award of the Nobel Peace Prize (2004) to an environmentalist, for the first time, came as a landmark decision, showing increasing global concern towards environmental issues and recognition to efforts being made for environmental conservation and protection.

NOBEL PEACE PRIZE, 2004 AND 2007 FOR ENVIRONMENTALISTS

The 2004 Nobel Peace Prize was awarded to Kenyan Environmentalist **Wangari Maathai** for her contribution to sustainable development, democracy and peace. This is the greatest recognition given to the cause of environment at international level. The Norwegian Nobel Committee, while awarding the prize, expressed the views ***"Peace on Earth depends on our ability to secure our living Environment".***

Wangari Maathai

Maathai, Kenya's Deputy Environment Minister is the founder of Kenya based **Green Belt Movement.** This movement comprising mainly of women has planted about 30 million trees across Africa. This has helped in slowing desertification, preserving forest habitats for wildlife and food for future generations and has helped combat poverty.

Maathai has given a beautiful slogan ***"When we plant new trees, we plant the seeds of peace."***

Nobel peace prize, 2007 was awarded jointly to Intergovernmental Panel on Climate Change (IPCC) headed by Indian Environmentalist Dr. R.K. Pachauri, and former US vice-president Al Gore. IPCC, the UN body comprising of 3,000 experts from various fields is an authority on global warming and its impacts. The award to IPCC is in appreciation of its efforts to build up and disseminate greater knowledge about man-made climate change and to lay the foundation for the measures that are needed to counteract such change. Al Gore is "probably the single individual who has done most to create greater world-wide understanding to the measures that need to be adopted," observed the Norwegian Nobel Committee while naming the joint winner of the award.

Former US vice-president Al Gore

R.K. Pachauri

(*b*) **Public Awareness for Environment**

The goals of sustainable development cannot be achieved by any government at its own level until the public has a participatory role in it. Public participation is possible only when the public is aware about the ecological and environmental issues.

The public has to be educated about the fact that if we are degrading our environment we are actually harming our own selves. This is because we are a part of the complex network of environment where every component is linked up. It is all the more important to educate the people that sometimes the adverse impact of environment are not experienced until a threshold is reached. So we may be caught unawares by a disaster.

A drive by the government to ban the littering of polythene cannot be successful until the public understands the environmental implications of the same. The public has to be made aware that by littering polythene, we are not only damaging the environment, but posing serious threat to our health.

There is a Chinese proverb ***"If you plan for one year, plant rice, if you plan for 10 years, plant trees and if you plan for 100 years, educate people."*** If we want to protect and manage our planet earth on sustainable basis, we have no other option but to make all persons environmentally educated.

(*c*) **Role of Contemporary Indian Environmentalists in Environmental Awareness**

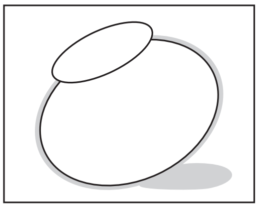
In our country, efforts to raise environmental awareness have been initiated, and several landmark judgements related to environmental litigations have highlighted the importance of this subject to general public. Two noted personalities who need a mention here, are Justice Kuldeep Singh, known popularly as ***the green judge*** and Sh. M.C. Mehta, ***the green advocate,*** who have immensely contributed to the cause of environment.

In 1991, the Supreme Court of our country issued directives to make all curricula environment-oriented. This directive was, in fact, in response to a Public Interest Litigation (PIL) filed by *M.C. Mehta vs. Union of India (1988)* that prompted the apex court to give a mandate for creating environmental awareness among all citizens of India. Based on the judgement, Environmental Studies is being taught as a compulsory course to all students.

There are some environmentalists in the present time who have made a mark in our country through environmental activism. Sh. Sunderlal Bahuguna, known for his 'Chipko movement' and 'Tehri Bachao Andolan', Smt. Medha Patkar and Ms. Arundhati Roy known for their 'Narmada Bachao Andolan', the Magsaysay awardee Sh. Rajender Singh known for his water conservation efforts are some such contemporary figures. Salim Ali is a renowned ornithologist, famous for his work on Indian birds. In modern India, our late Prime Minister Mrs. Indira Gandhi was instrumental in introducing the concept of environmental protection in the Constitution of India as a fundamental duty while Mrs. Maneka Gandhi, formerly environment minister, has worked a lot for the cause of wildlife protection. Citizens report on environment was first published by late Sh. Anil Aggarwal, the founder Chairman of Centre for Science & Environment. Even with many such key persons leading the cause to environment, India is yet to achieve a lot in this field.

(*d*) **Role of Government**

Concept of Ecomark: In order to increase consumer awareness about environment, the Government of India has introduced a scheme of eco-labelling of consumer products as 'Ecomark' in 1991. It is an 'earthen pitcher'–a symbol of eco-friendliness and our traditional heritage. A product that is made, used or disposed off in a harmless manner is called eco-friendly and is awarded this eco-mark.



Ecomark of India

In a drive to disseminate environmental awareness **'Eco-Clubs'** for children and **'Eco-task force'** for army men have also been launched by the government.

Today everybody talks of environment, but only a few have clear ideas about what needs to be done and still fewer people have the actual experience or expertise in the field. Unfortunately, environmental awareness campaigns have very often been exploited for political propaganda rather than being an integral part of our educational programmes in theory and practice. "*Environment" is very wrongly taken as a "fashion" by all walks of life, hardly realizing that it is our "real-life-situation" and our sustenance and security are at stake*.

To sum up, it may be said that it is absolutely essential to create environmental awareness because:

(*i*) Environment belongs to all and participation of masses is a must for successful implementation of environmental protection plans.

(*ii*) Living in a technologically developing society, our lifestyles and attitudes have become self-oriented. Environmental awareness is needed to change the mindset of modern society for an earth-oriented approach.

(*iii*) There is a need to make the public environmentally aware of the serious health impacts of environmental pollution and their right to live in a clean and healthy environment.

(*iv*) There is an urgent need to create awareness amongst people that we have no other option but to follow sustainability principles. Only then life of mankind on this earth would be secure and our future generations would be safe.

Henry D. Thoreau had rightly said ***"What's the use of a beautiful house if you don't have a decent planet to put it on?"*** Even if we begin today, the restoration is expected in the next 40–50 years.

1.5 CONCEPT OF ECOLOGY AND ECOSYSTEM

Various kinds of life supporting systems like the forests, grasslands, oceans, lakes, rivers, mountains, deserts and estuaries show wide variations in their structural composition and functions. However, they all are alike in the fact that they consist of living entities interacting with their surroundings exchanging matter and energy. How do these different units like a hot desert, a dense evergreen forest, the Antarctic Sea or a shallow pond differ in the type of their flora and fauna, how do they derive their energy and nutrients to live together, how do they influence each other and regulate their stability are the questions that are answered by Ecology.

The term Ecology was coined by Earnst Haeckel in 1869. It is derived from the Greek words *Oikos*- home + *logos*- study. So **ecology deals with the study of organisms in their natural home**

interacting with their surroundings. The surroundings or environment consists of other living organisms (biotic) and physical (abiotic) components. Modern ecologists believe that an adequate definition of ecology must specify some unit of study and one such basic unit described by Tansley (1935) was ecosystem. **An ecosystem is a self-regulating group of biotic communities of species interacting with one another and with their non-living environment exchanging energy and matter.** Now **ecology** is often defined as **"the study of ecosystems".**

An ecosystem is an integrated unit consisting of interacting plants, animals and micro-organisms whose survival depends upon the maintenance and regulation of their biotic and abiotic structures and functions. The ecosystem is thus, a unit or a system which is composed of a number of sub-units, that are all directly or indirectly linked with each other. They may be freely exchanging energy and matter from outside—an *open ecosystem* or may be isolated from outside in term of exchange of matter—a *closed ecosystem.*

Ecosystems show large variations in their size, structure, composition etc. However, all the ecosystems are characterized by certain basic structural and functional features which are common. Composition and organization of biological communities and abiotic components constitute the structure of an ecosystem. Thus, ecosystems have basically two types of components, the biotic and abiotic, as described below:

(*a*) **BIOTIC COMPONENTS:** Different living organisms constitute the biotic component of an ecosystem and belong to the following categories:

(*i*) **Producers:** These are mainly producing food themselves *e.g.,* Green plants produce food by photosynthesis in the presence of sunlight from raw materials like water and carbon dioxide. They are known as *photo-autotrophs* (auto = self, photo = light, troph = food).

There are some *chemo-autotrophs*, which are a group of bacteria, producing their food from oxidation of certain chemicals. *e.g.* sulphur bacteria.

(*ii*) **Consumers:** These organisms get their food by feeding on other organisms. They are of the following types:

- Herbivores—which feed on plants *e.g.* rabbit, insect.
- Carnivores—which feed on herbivores as secondary carnivores (*e.g.,* frog, small fish) or tertiary carnivores (*e.g.,* snake, big fish), which feed on other consumers.
- Omnivores—which feed on both plants and animals *e.g.,* humans, rats, many birds.
- Detritivores—which feed on dead organisms *e.g.,* earth worm, crab, ants.

(*iii*) **Decomposers:** These are micro-organisms which break down organic matter into inorganic compounds and in this process they derive their nutrition. They play a very important role in converting the essential nutrients from unavailable organic form to free inorganic form that is available for use by plants *e.g.,* bacteria, fungi.

(*b*) **ABIOTIC COMPONENTS:** Various physico-chemical components of the ecosystem constitute the abiotic structure:

(*i*) Physical components include sunlight, solar intensity, rainfall, temperature, wind speed and direction, water availability, soil texture etc.

(*ii*) Chemical components include major essential nutrients like C, N, P, K, H_2, O_2, S etc. and micronutrients like Fe, Mo, Zn, Cu etc., salts and toxic substances like pesticides.

These physico-chemical factors of water, air and soil play an important role in ecosystem functioning.

Every ecosystem performs the following important functions:

(*i*) **It has different food chains and food webs.** Food chain is the sequence of eating and being eaten. *e.g.,*

Grass → Grasshopper → Frog → Snake → Hawk

Phytoplanktons (water-algae) → water fleas → small fish → large fish (Tuna)

These are known as **grazing food chain—**which start with green plants and culminate with carnivores.

Another type is **detritus food chain—**which starts with dead organic matter. *e.g.,*

Leaf litter in a forest → Fungi → bacteria

Food chains are generally found to be interlinked and inter-woven as a network and known as **Food Web.** There are several options of eating and being eaten in a food web. Hence these are more stable.

(*ii*) **There is uni-directional flow of energy in an ecosystem.** It flows from sun and then after being captured by primary producers (green plants), flows through the food chain or food web, following the laws of thermodynamics. At every successive step in the food-chain, there is huge loss of about 90% of the energy in different processes (respiration, excretion, locomotion etc.) and only 10% moves to next level (Ten per cent law of energy flow).

(*iii*) **Nutrients (Materials) in an ecosystem move in a cyclic manner.** The cycling of nutrients takes place between the biotic and abiotic components, hence known as biogeochemical cycles (bio = living, geo = earth, chemical = nutrients).

(*iv*) **Every ecosystem functions to produce and sustain some primary production (plant biomass) and secondary production (animal biomass).**

(*v*) **Every ecosystem regulates and maintains itself and resists any stresses or disturbances up to a certain limit.** This self regulation or control system is known as **cybernetic system**.

1.6 BALANCED ECOSYSTEM

Ecosystems have a unique property of self-regulation. The ecosystem comprising various sub-components of biotic and abiotic nature, which are inter-linked and inter-dependent, have an inherent property to resist change. That means, the ecosystems have a property to tolerate external disturbance or stress. This property is known as **homeostasis.** The ecosystems have a definite structure comprised of certain types of living organisms, which have a definite place and role in the ecosystem, as defined by their position in the food-web. Together, in interaction with the abiotic components, these ecosystems perform the functions of energy flow and material cycling, and finally give a desired output in the form of productivity. Every ecosystem can operate within a range of conditions, depending upon its **homeostasis** (capacity to resist change). Within its homeostatic plateau, the ecosystem has the potential to trigger certain feedback mechanisms which help in maintaining the ecosystem functioning by countering the disturbances. Such **deviation-counteracting feedbacks** are known as **negative feedback mechanisms.** Such feedback loops help in maintaining the ecological balance of the ecosystem.

A balanced ecosystem has basic biotic components which have evolved with time to suit the environmental conditions. The flow of energy and cycling of nutrients take place in a definite pattern in such an ecosystem, under a set of physical environment.

However, as the outside disturbance or stress increases beyond a certain limit (exceeding the homeostatic plateau of the ecosystem), the balance of the ecosystem is disrupted. This is because now another type of feedback mechanisms, which are **deviation accelerating mechanisms** start operating. Such feedbacks are called **positive feedback** mechanisms, which further increase the disturbances caused by the external stress and thus take the ecosystem away from its optimal conditions, finally leading to collapse of the system.

Figure 1.2 depicts the control system of a balanced ecosystem within a range.

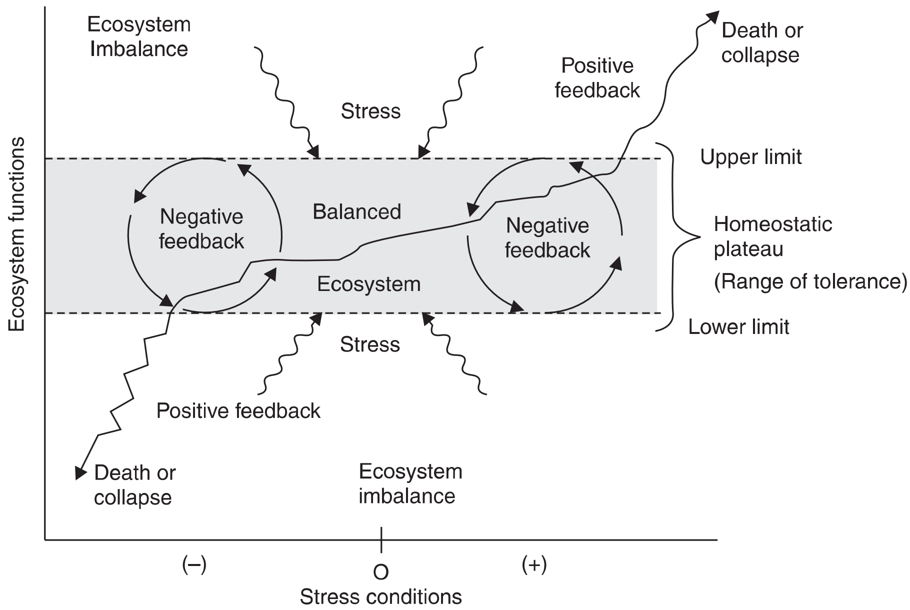


Fig. 1.2. Ecosystem regulation by homeostasis. On application of a stress, the negative feedback mechanisms start operating, trying to counter the stress and regulate the balance of the system but beyond the homeostatic plateau, positive feedback starts which further accelerate the stress effects causing ecosystem imbalance.

To understand the concept we can consider the following example.

Carbon dioxide is required by green plants to manufacture their food during photosynthesis and the food produced by green plants is actually the base of food chains, energy flow and material cycles. The ecosystems have an excellent balance of regulating the levels of carbon dioxide through carbon cycle, where all living organisms produce CO_2 during respiration and the green plants use them up during photosynthesis, liberating oxygen. Upto certain limits, increase in CO_2 concentrations can help in improving production by green plants. But beyond a limit, the increased CO_2 will cause an imbalance in the ecosystem triggering various harmful positive feedbacks. As a result, several adverse environmental impacts occur including global warming, changing rainfall patterns, crop insecurity, storms, flooding, emergence of new types of pests—all leading to degradation of the ecosystem.

I. QUESTIONS

1. What is the need for studying environmental issues?
2. What is the scope of environmental education?
3. How would environmental awareness help to protect our environment?
4. Define environment.
5. How does urban environment differ from rural environment?
6. What is meant by 'Ecomark'?
7. How do we apply Mathematics and Engineering aspects to environmental studies?
8. What is green marketing?
9. Define ecology and ecosystems.
10. What are the biotic and abiotic components of an ecosystem?
11. What are food chains and food webs? Give examples and discuss their significance.
12. What are the types of feedback mechanisms regulating an ecosystem?
13. What are the functions of an ecosystem?

II. OBJECTIVE TYPE QUESTIONS

(A) FILL IN THE BLANKS

1. The term 'Environment' has been derived from the French word .... which means to encircle or surround.
2. The United Nations Conference on Environment and Development (Earth Summit) was held at .... in ....
3. The World Summit on Sustainable Development was held at .... in ....
4. Sunderlal Bahuguna is associated with popular environmental movements, ... and ....
5. Mr. .... filed PIL (Public Interest Litigation) for creating environmental awareness among all citizens of India.
6. ...... was awarded the Nobel Peace Prize in 2004 for her contribution towards environmental conservation.
7. ...... is popularly known as green judge in India.
8. World Environment Day is celebrated on ....

9. Environment friendly products are given ISO certification called ISO ....
10. Ecomark of our country is ....
11. The term 'ecology' was coined by ...... and 'ecosystem' was given by...... .
12. The sequence of eating and being eaten in an ecosystem is called a ....
13. Microbes (bacteria) capable of producing organic food by oxidation of certain chemicals are known as ....
14. Movement of nutrients in an ecosystem is cyclic and flow of energy is ....

(B) CHOOSE THE CORRECT ANSWER

1. Environment involves the interaction between
(*a*) Air, water and living organisms
(*b*) Air, water, land and living organisms
(*c*) Air, water, land, materials and living organisms
(*d*) None of these.

2. Which of the following involves children in environmental awareness?
(*a*) Ecomark (*b*) Ecoclub
(*c*) Eco task force (*d*) All of these.

3. Which of the following is NOT a correct pair?
(*a*) World Forest Day—March 21
(*b*) Earth Day—June 5
(*c*) World Population Day—July 11
(*d*) World Day for Water—March 22.

4. Which of the following is an incorrect pair?
(*a*) R.K. Pachauri—Nobel Peace Prize
(*b*) M.C. Mehta—Green Advocate
(*c*) Rajender Singh—Water Man of India
(*d*) Medha Patekar—Chipko Movement.

5. The organisms feeding on dead organisms are called
(*a*) Carnivores
(*b*) Decomposers
(*c*) Detritivores
(*d*) Omnivores.

6. Which of the following is not an abiotic factor?
(*a*) Solar radiations (*b*) Bacteria
(*c*) Soil texture (*d*) Carbon.

(C) WRITE TRUE OR FALSE

1. Only the physical world surrounding us is our environment. (True/False)
2. Ozone week is observed during September 16–23. (True/False)
3. Ecology may be defined as "the study of ecosystems." (True/False)

4. Ecosystems that freely exchange energy and matter from outside are known as closed ecosystems. (True/False)
5. An ecosystem is a group of biotic communities interacting with one another but without exchanging energy and matter with the non-living environment. (True/False)
6. Detritivores are also known as saprotrophs. (True/False)

   Biotic and abiotic components of an ecosystem influence each other and are linked through energy flow and matter cycling. (True/False)
7. Food webs provide less stability to an ecosystem as compared to linear food chain. (True/False)
8. If we want to maintain the ecological balance, we should try to contribute to positive feedback mechanisms. (True/False)
9. Positive feedback mechanisms tend to take the system away from its optimal conditions. (True/False)
10. The range of tolerance of an ecosystem to a particular stress is best described by its homeostatic plateau. (True/False)